



National Institute of Business Management

Mobile Application Development (Project: SnapShop)

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Course	:	HND-SE
Group Number	:	01
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Project Description

In everyday life, professionals, technicians, and general users frequently interact with physical products, specialized tools, and hardware equipment. A common friction point occurs when a user needs to purchase or replace a specific item but lacks the ability to describe it or does not know its exact model name and where it is sold.

SnapShop is a utility-driven Android application designed to bridge this gap between physical observation, sourcing and purchasing. By taking a photo of any ambiguous item or tool, the app automates the complex pipeline of object isolation, multi-modal visual recognition, natural language query distillation, and real-time market indexing to present the user with direct, verified purchase locations.

- **Project Technical Brief**

Product Name	SnapShop
Architecture	Clean Architecture
Project Dev Lang.	Kotlin + Jetpack Compose
Target Android API	API Level 28+

- **Target Audience**

- **Technicians, Engineers:** Who need to quickly identify and replace specific mechanical parts, specialized tools, or electronics encountered on-site.
- **General Consumers:** Who spot interesting lifestyle goods, office equipment, or gadgets in daily life and want to check pricing and local availability instantly.
- **DIY Enthusiasts:** Who work with various materials and hardware components but do not always know the exact industry names for them.

- **Unique Selling Point**

Instead of forcing users to find what's in front of them, guess the product or scroll endlessly through generic search engine results, **SnapShop** provides immediate clarity. It transforms raw, chaotic image capture into structured, actionable purchasing options within seconds. The app eliminates descriptive guesswork, minimizes shopping friction, and preserves user context by mapping and saving their physical discovery history.

Feature List & Prioritization

Below are the core features provided by the SnapShop app, they are sorted in a manner based on how the app works step by step:

- **Core Features**

1. **In-App Camera Capture:** A native camera interface that lets users point and shoot to snap high-quality photos of items or tools with a simple shutter button press.
2. **Smart Background Removal:** Uses on-device Google ML Kit Object Detection to instantly isolate the main subject and crop it into a clean square, stripping away background clutter and visual noise.
3. **Visual Product Recognition:** Ability to scan the captured image for visual web indexes and retrieve the top 5 most accurate product name matches.
4. **E-Commerce Store & Link Indexing:** Feeds the identified product name back into product search to pull active online marketplace listings, store names, and direct merchant checkout links.
5. **Local History & Caching:** Stores every successful scan, and product listing link locally in a SQLite database to provide fast offline access and act as a performance shield.
6. **User Identity & State Management:** Handles secure user sign-up and sign-in flows using Firebase Authentication to keep user profiles and personal history secure.
7. **Cloud Synchronization:** Automatically backs up the local database history array to a remote instance, ensuring user data is synchronized across different devices.

To ensure a structured development cycle, the features are prioritized into three distinct tiers based on core system dependencies.

- **Features Implementation Prioritization**

1. Core Identification Pipeline (Highest Priority)
 - **Camera Integration:** High-performance, in-app camera capture with shutter controls using Android modern camera APIs.
 - **On-Device Subject Isolation:** Real-time object detection to automatically crop the target item into a clean square, stripping out background noise before processing.

- **Multi-Stage Search:** Sequence-dependent orchestration search via image hosting upload, SerpAPI Google Lens endpoints, and Gemini API inference blocks.
 - **Query Distillation Engine:** Feeding raw visual string metadata to an LLM configuration to synthesize a clean, unified English search term (Product Name).
 - **Market Indexing:** Passing the refined product term to dedicated product search APIs to extract direct web links, store names, and active pricing.
2. Data Persistence & Session Management (Medium Priority)
 - **Local Caching (SQLite):** Offline access to search queries and results to ensure users can access their scan records instantaneously without triggering redundant, expensive external API calls.
 - **Authentication:** Implementing secure user sign-in to save scan records and data.
 - **Cloud Sync:** Backing up local history data arrays to a remote database so a user's history persists if they switch physical devices.
 3. Deployment & Administrative Tooling (Lowest Priority)
 - **GitHub Repository & Documentation:** Packaging the entire clean architecture pattern, dependency injection modules, and instructions into a public repository for technical presentation.

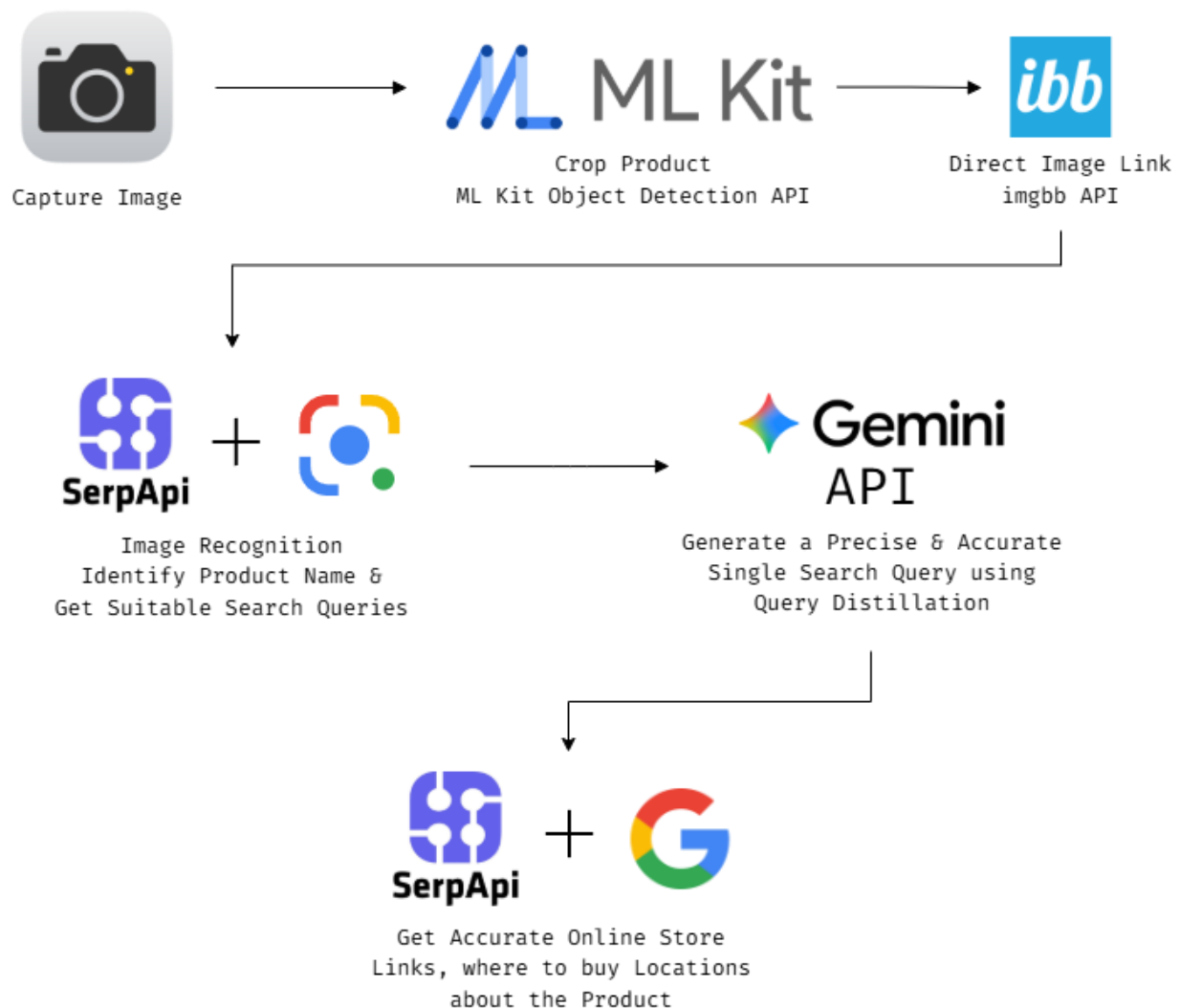
Technical Workflow

The app is planning to build using a modern Android architecture pattern, utilizing official Jetpack libraries and specific cloud endpoints to execute the pipeline effectively and efficiently, below is how the app works in technical perspective.

1. **Camera Alignment:** The user aims the device camera at the target tool or product in their physical environment.
2. **Image Capture:** Tapping the shutter button triggers the in-app camera to capture a high-resolution frame of the item.
3. **On-Device Cropping:** Google ML Kit Object Detection processes the frame to calculate a bounding box and crop the subject, eliminating background noise.
4. **CDN Upload:** The cropped image is uploaded to Imgbb to generate a direct, public image URL required by the backend search engines.

5. **Visual Analysis (SerpAPI):** The image URL is parsed by the SerpAPI Google Lens endpoint to extract raw title metadata from the top 5 visual matches.
6. **Query Distillation (Gemini API):** The 5 raw titles are sent to Gemini 3.1 Flash-Lite to filter out e-commerce clutter, translate foreign terms, and output a single, optimized English product name.
7. **Market Indexing (SerpAPI):** The distilled product name is passed to SerpAPI Google Shopping to fetch live retail links, store names, and active pricing.

Diagram of How the Process Works



Development Tools & Libraries

Below are the tools, utilities and libraries we intend to use.

1. Core Platform & UI Layer

- **Kotlin:** The primary programming language, leveraging Coroutines and Asynchronous Flows for non-blocking network and database execution.
- **Jetpack Compose:** Used to design a highly responsive, modern, declarative User Interface utilizing Material 3 components.
- **Android SDK (API Level 28+):** Ensures compatibility with modern Android operating systems while preserving access to native machine learning frameworks.

2. On-Device Image Processing

- **Google ML Kit (Object Detection & Tracking API):** Used on device to isolate the primary subject inside the camera viewport and calculate a precise bounding box crop, removing environmental noise.

3. Networking & Infrastructure

- **OkHttp3:** The networking standard used to configure REST API requests, manage request headers and serialize JSON data payloads.
- **Imgbb API:** Serves as the intermediary CDN cloud storage engine to host the cropped product images dynamically, generating a direct public URL string required by external processing platforms.

4. Cloud Reasoning & Intelligence APIs

- **SerpAPI (Google Lens API):** Processes the public image URL to perform deep reverse-image tracking and returns the top 5 highly matching structural product records.
- **Google Gemini 3.1 Flash-Lite API:** Utilizes the Structured Outputs (response_schema) feature via standard HTTP REST payloads. It runs a natural language query distillation loop with low temperature configurations (0.1) to reliably return a clean e-commerce text query wrapped in strict JSON format.
- **SerpAPI (Google Product Search API):** Ingests the polished text query from Gemini to pull active live store configurations and e-commerce shopping links.

5. Storage Persistence & Identity

- **Room Database (SQLite Abstraction):** Handles local relational data storage on the device, functioning as a performance shield to cache past scans and save system quotas.
- **Firebase Authentication:** Controls secure account creation, user state monitoring, and credential management.
- **Cloud Firestore:** Serves as the highly scalable cloud database to automatically back up and synchronize local search history configurations across unique user nodes.

Contribution Planning

Below is how we divided each crucial development step and roles among our team members, expertise for each area is handled and managed by relevant individuals,

DevOps & Project Management

MAHNDSE261F-001 D.P.D Madushan	• Project Management, Version controlling, Deadlines handling • Critical Business Logic Management, • Review and Implementations of Critical Areas
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Business Logic & Internal Methods

MAHNDSE261F-005 K.L. Thineth Geevinda	• Business Logic Implementation • Utility methods and Internal logic implementation
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UI Design

MAHNDSE261F-006 K.K.N Naveesha	• UI Design Mockups, Review in development User Interfaces • Maintain the UI for best visual integrity
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UX Design

MAHNDSE261F-010 G.A Disanayaka	• Design and Plan user experience inside the app. • Manage and review how the app should behave after launch
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Generally, every team member is responsible for implementing features as part of the team. The above is a brief overview of their specific roles, but that does not mean they are only accountable for that part or feature. Every member is dedicated to implementing the full app as a team.